

Trees of the same status sequence

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Abstract

Suppose T is a tree and there exist a pair of distinct non-endvertices x, y in T with the same status. If T' is an equal-component-transferred tree of T at x and y , then T' and T have the same status sequence.

Keywords: status, status sequence, tree, component-transferred tree, equal-component-transferred tree

1 Introduction

Let G be a finite simple connected graph without loops. For a given vertex x of G , the *status* of x , denoted by $s_G(x)$, is defined by $s_G(x) = \sum_{y \in V(G)} d_G(y, x)$, where $d_G(y, x)$ is the distance between y and x . The *status sequence* of G is the list of the statuses of all the vertices of G arranged in nondecreasing order. It is well known that non-isomorphic graphs may have the same status sequence. A spider is a tree of which one and only one vertex has degree exceeding 2. Slater [2] provided infinite pairs of non-isomorphic trees with the same status sequence. A spider, however, is uniquely determined in trees by its status sequence. Shang and Lin [1] showed that whenever a tree T and a spider S have the same status sequence, then $T \cong S$. Motivated by these facts, the following problem is considered: For a given tree T , can we construct a tree T' such that $T' \not\cong T$ and T' and T have the same status sequence? Now begin with the following definitions. Let T be a tree and x, y a pair of distinct non-endvertices in T . Let C be a component of $T - \{x, y\}$. If for every $t \in V(C)$, $d_T(t, x) < d_T(t, y)$, then C is called a *x-adjacent component* of $T - \{x, y\}$. Similarly, if for every $t \in V(C)$, $d_T(t, y) < d_T(t, x)$, then C is called a *y-adjacent component* of $T - \{x, y\}$. If C is neither a *x-adjacent component* nor a *y-adjacent component* of $T - \{x, y\}$, then C is called

a *x, y-adjacent component* of $T - \{x, y\}$. Let $\deg_T v$ denote the degree of the vertex v in T . Obviously, the graph $T - \{x, y\}$ have $\deg_T x - 1$ *x-adjacent components* and $\deg_T y - 1$ *y-adjacent components*, respectively. And if x and y are adjacent in T , then $T - \{x, y\}$ has no *x, y-adjacent component*; if x and y are not adjacent in T , then $T - \{x, y\}$ has one and the only one *x, y-adjacent component*. Let $A_1, A_2, \dots, A_p$ be some *x-adjacent components* and $B_1, B_2, \dots, B_q$ some *y-adjacent components* of $T - \{x, y\}$, respectively, where $1 \leq p \leq \deg_T x - 1$ and $1 \leq q \leq \deg_T y - 1$. Let $x_i \in V(A_i)$ and $y_j \in V(B_j)$ such that $x_i x, y_j y \in E(T)$ for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, q$. Let F be the graph obtained by deleting all the edges $x_1 x, x_2 x, \dots, x_p x$ and $y_1 y, y_2 y, \dots, y_q y$ from T . And let T' be the tree obtained by adding the edges $x_1 y, x_2 y, \dots, x_p y$ and $y_1 x, y_2 x, \dots, y_q x$ to F . Then T' is called a *component-transferred tree* of T at x and y . And the *x-adjacent components* $A_1, A_2, \dots, A_p$ and the *y-adjacent components* $B_1, B_2, \dots, B_q$ of $T - \{x, y\}$ are called the *transfer components* from T to T' at x and y . Furthermore, if $\sum_{i=1}^p |V(A_i)| = \sum_{j=1}^q |V(B_j)|$, then T' is called an *equal-component-transferred tree* of T at x and y . The main result of this paper is that if T' is an equal-component-transferred tree of a tree T at a pair of distinct non-endvertices x and y in T with $s_T(x) = s_T(y)$, then T' and T have the same status sequence. Note that T' and T are not necessarily non-isomorphic. Fig. 1 below shows that T_2 is an equal-component-transferred tree of T_1 at x and y , where x and y are the vertices with $s_{T_1}(x) = s_{T_1}(y) = 32$. Both T_1 and T_2 have status sequence $\{26, 26, 28, 32, 32, 36, 36, 40, 42, 42, 42, 50\}$ and $T_1 \not\cong T_2$. In Fig. 1 the number beside each vertex of T_1 and T_2 is the status of the vertex.

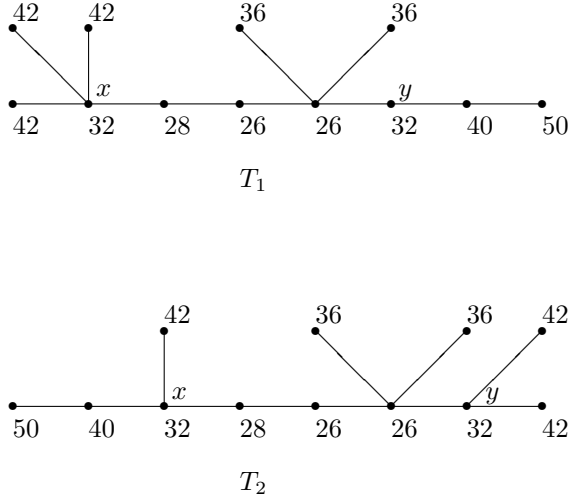


Fig. 1

2 Main Results

This section begins with the following lemma.

Lemma 2.1 *Let T be a tree and x, y a pair of distinct non-endvertices in T with $s_T(x) = s_T(y)$. If T' is an equal-component-transferred tree of T at x and y , then $s_{T'}(x) = s_T(x)$ and $s_{T'}(y) = s_T(y)$.*

Proof. Let $A_1, A_2, \dots, A_{deg_T x - 1}$ be the x -adjacent components and $B_1, B_2, \dots, B_{deg_T y - 1}$ the y -adjacent components of $T - \{x, y\}$, respectively. Without loss of generality, let $A_1, A_2, \dots, A_p$ and $B_1, B_2, \dots, B_q$ be the transfer components from T to T' at x and y with $\sum_{i=1}^p |V(A_i)| = \sum_{j=1}^q |V(B_j)|$, where $1 \leq p \leq deg_T x - 1$ and $1 \leq q \leq deg_T y - 1$. Note that $V(T) = V(T')$ and A_i is a y -adjacent component of $T' - \{x, y\}$ for $i = 1, 2, \dots, p$, and B_j is a x -adjacent component of $T' - \{x, y\}$ for $j = 1, 2, \dots, q$. We have $d_T(t, x) = d_{T'}(t, y)$ for every $t \in \bigcup_{i=1}^p V(A_i)$ and $d_T(t, y) = d_{T'}(t, x)$ for every $t \in \bigcup_{j=1}^q V(B_j)$. And $d_T(t, x) = d_{T'}(t, x)$ for every $t \in V(T) - \bigcup_{i=1}^p V(A_i) - \bigcup_{j=1}^q V(B_j)$. Then

$$\begin{aligned} & s_T(x) \\ &= \sum_{t \in V(T)} d_T(t, x) \\ &= \sum_{t \in \bigcup_{i=1}^p V(A_i)} d_T(t, x) + \sum_{t \in \bigcup_{j=1}^q V(B_j)} d_T(t, x) \end{aligned}$$

$$\begin{aligned} &+ \sum_{t \in V(T) - \bigcup_{i=1}^p V(A_i) - \bigcup_{j=1}^q V(B_j)} d_T(t, x) \\ &= \sum_{t \in \bigcup_{i=1}^p V(A_i)} (d_T(t, x) + d_T(y, x)) + \\ &\quad \sum_{t \in \bigcup_{j=1}^q V(B_j)} (d_T(t, x) - d_T(y, x)) + \\ &\quad \sum_{t \in V(T) - \bigcup_{i=1}^p V(A_i) - \bigcup_{j=1}^q V(B_j)} d_T(t, x) \\ &= \sum_{t \in \bigcup_{i=1}^p V(A_i)} (d_{T'}(t, y) + d_{T'}(y, x)) + \\ &\quad \sum_{t \in \bigcup_{j=1}^q V(B_j)} d_{T'}(t, y) + \\ &\quad \sum_{t \in V(T) - \bigcup_{i=1}^p V(A_i) - \bigcup_{j=1}^q V(B_j)} d_{T'}(t, x) \\ &= \sum_{t \in \bigcup_{i=1}^p V(A_i)} d_{T'}(t, x) + \sum_{t \in \bigcup_{j=1}^q V(B_j)} d_{T'}(t, x) \\ &\quad + \sum_{t \in V(T) - \bigcup_{i=1}^p V(A_i) - \bigcup_{j=1}^q V(B_j)} d_{T'}(t, x) \\ &= \sum_{t \in V(T)} d_{T'}(t, x) \\ &= s_{T'}(x). \end{aligned}$$

Similarly, $s_T(y) = s_{T'}(y)$. $\square$

Lemma 2.2 (Remark 2.4 [1])

Let $x_1 x_2 x_3 \dots x_k$ ($k \geq 2$) be a path in a tree T . Suppose that $T_1, T_2, \dots, T_k$ are components of $T - \{x_1 x_2, x_2 x_3, \dots, x_{k-1} x_k\}$ with $x_i \in V(T_i)$ for $i = 1, 2, \dots, k$. Then

$$s_T(x_1) - s_T(x_k) = \sum_{i=1}^k (-k - 1 + 2i) |V(T_i)|.$$

$\square$

Theorem 2.3 *Let T be a tree and x, y a pair of distinct non-endvertices in T with $s_T(x) = s_T(y)$. If T' is an equal-component-transferred tree of T at x and y , then T' and T have the same status sequence.*

Proof. Let $A_1, A_2, \dots, A_{deg_T x - 1}$ be the x -adjacent components and $B_1, B_2, \dots, B_{deg_T y - 1}$ the y -adjacent components of $T - \{x, y\}$, respectively. W.l.o.g., let $A_1, A_2, \dots, A_p$ and $B_1, B_2, \dots, B_q$ be the transfer components from T to T' at x and y with $\sum_{i=1}^p |V(A_i)| = \sum_{j=1}^q |V(B_j)|$, where $1 \leq p \leq deg_T x - 1$ and $1 \leq q \leq deg_T y - 1$. By Lemma 2.1, $s_{T'}(x) = s_T(x)$ and $s_{T'}(y) = s_T(y)$. Let $V(T) = \{x_1 =$

$x, x_2 = y, x_3, \dots, x_n\}$. Then $V(T') = \{x_1 = x, x_2 = y, x_3, \dots, x_n\}$. For $x_k \in V(T)$ where $k \geq 3$, we distinguish the following three cases and in each case show that $s_{T'}(x_k) = s_T(x_k)$. Case 1. $x_k \in \bigcup_{i=1}^{deg_T x - 1} V(A_i)$, Case 2. $x_k \notin (\bigcup_{i=1}^{deg_T x - 1} V(A_i)) \cup (\bigcup_{j=1}^{deg_T y - 1} V(B_j))$ and Case 3. $x_k \in \bigcup_{j=1}^{deg_T y - 1} V(B_j)$. In cases 1 and 2, consider the path $x_{t_1} x_{t_2} \dots x_{t_r}$ in T where $x_{t_1} = x$, $x_{t_r} = x_k$ and $r \geq 2$. Let $T_1, T_2, \dots, T_r$ be the components of $T - \{x_{t_1} x_{t_2}, x_{t_2} x_{t_3}, \dots, x_{t_{r-1}} x_{t_r}\}$ such that $x_{t_i} \in V(T_i)$ for $i = 1, 2, \dots, r$. By Lemma 2.2, $s_T(x) - s_T(x_k) = s_T(x_{t_1}) - s_T(x_{t_r}) = \sum_{i=1}^r (-r - 1 + 2i)|V(T_i)|$.

Case 1. $x_k \in \bigcup_{i=1}^{deg_T x - 1} V(A_i)$.

Now suppose that $x_k \in V(A_{i_0})$ for some $1 \leq i_0 \leq deg_T x - 1$.

Subcase 1.1. $p + 1 \leq i_0 \leq deg_T x - 1$.

Since $x_{t_1} x_{t_2} \dots x_{t_r}$ is also a path in T' . Let $T'_1, T'_2, \dots, T'_r$ be the components of $T' - \{x_{t_1} x_{t_2}, x_{t_2} x_{t_3}, \dots, x_{t_{r-1}} x_{t_r}\}$ with $x_{t_i} \in V(T'_i)$ for $i = 1, 2, \dots, r$. By Lemma 2.2 again, $s_{T'}(x) - s_{T'}(x_k) = s_{T'}(x_{t_1}) - s_{T'}(x_{t_r}) = \sum_{i=1}^r (-r - 1 + 2i)|V(T'_i)|$. Obviously, $V(T_1) = V(T) - V(A_{i_0}) = V(T'_1)$ and $V(T_i) = V(T'_i)$ for $i = 2, \dots, r$. Then $s_T(x_k) = s_T(x) + \sum_{i=1}^r (r + 1 - 2i)|V(T_i)| = s_{T'}(x) + \sum_{i=1}^r (r + 1 - 2i)|V(T'_i)| = s_{T'}(x_k)$.

Subcase 1.2. $1 \leq i_0 \leq p$.

Note that A_{i_0} is a x -adjacent component of $T - \{x, y\}$ and a y -adjacent component of $T' - \{x, y\}$. For the path $y x_{t_2} x_{t_3} \dots x_{t_r}$ in T' , let $T'_1, T'_2, \dots, T'_r$ be the components of $T' - \{y x_{t_2}, x_{t_2} x_{t_3}, \dots, x_{t_{r-1}} x_{t_r}\}$ with $y \in V(T'_1)$ and $x_{t_i} \in V(T'_i)$ for $i = 2, \dots, r$. By Lemma 2.2 again, $s_{T'}(y) - s_{T'}(x_k) = \sum_{i=1}^r (-r - 1 + 2i)|V(T'_i)|$. Obviously, $V(T_1) = V(T) - V(A_{i_0}) = V(T'_1)$ and $V(T_i) = V(T'_i)$ for $i = 2, \dots, r$. Then $s_T(x_k) = s_T(x) + \sum_{i=1}^r (r + 1 - 2i)|V(T_i)| = s_{T'}(y) + \sum_{i=1}^r (r + 1 - 2i)|V(T'_i)| = s_{T'}(x_k)$ for $s_T(x) = s_T(y) = s_{T'}(y)$ by Lemma 2.1.

Case 2.

$x_k \notin (\bigcup_{i=1}^{deg_T x - 1} V(A_i)) \cup (\bigcup_{j=1}^{deg_T y - 1} V(B_j))$.

In this case, x_k is in the x, y -adjacent component of $T - \{x, y\}$. We see that $V(T_1) = \{x\} \cup (\bigcup_{i=1}^{deg_T x - 1} V(A_i))$. Let T_s be the component among $T_1, T_2, \dots, T_r$ such that $y \in V(T_s)$. Then $(\bigcup_{j=1}^{deg_T y - 1} V(B_j)) \subset V(T_s)$. Since T' is an equal-component-transferred tree of T at x and y with transfer components $A_1, A_2, \dots, A_p$ and $B_1, B_2, \dots, B_q$, it is obvious that $x_{t_1} x_{t_2} \dots x_{t_r}$ is also a path in T' , where $x_{t_1} = x$ and $x_{t_r} = x_k$. Let $T'_1, T'_2, \dots, T'_r$ be the components of $T' - \{x_{t_1} x_{t_2}, x_{t_2} x_{t_3}, \dots, x_{t_{r-1}} x_{t_r}\}$ with $x_{t_i} \in V(T'_i)$ for $i = 1, 2, \dots, r$. Then by Lemma 2.2 again, $s_{T'}(x) -$

$s_{T'}(x_k) = \sum_{i=1}^r (-r - 1 + 2i)|V(T'_i)|$. Obviously, $V(T'_1) = \{x\} \cup (\bigcup_{i=p+1}^{deg_T x - 1} V(A_i)) \cup (\bigcup_{j=1}^q V(B_j))$, $V(T'_s) = (V(T_s) - (\bigcup_{j=1}^q V(B_j))) \cup (\bigcup_{i=1}^p V(A_i))$ and $V(T_i) = V(T'_i)$ for $i \neq s$ and $2 \leq i \leq r$. Then $|V(T_1)| = 1 + \sum_{i=1}^{deg_T x - 1} |V(A_i)| = 1 + \sum_{i=1}^p |V(A_i)| + \sum_{i=p+1}^{deg_T x - 1} |V(A_i)| = 1 + \sum_{j=1}^q |V(B_j)| + \sum_{i=p+1}^{deg_T x - 1} |V(A_i)| = |V(T'_1)|$ and $|V(T_s)| = |V(T'_s)| - \sum_{j=1}^q |V(B_j)| + \sum_{i=1}^p |V(A_i)| = |V(T'_s)|$. Hence $s_T(x_k) = s_T(x) + \sum_{i=1}^r (r + 1 - 2i)|V(T_i)| = s_{T'}(x) + \sum_{i=1}^r (r + 1 - 2i)|V(T'_i)| = s_{T'}(x_k)$.

Case 3. $x_k \in \bigcup_{j=1}^{deg_T y - 1} V(B_j)$.

In this case consider the path $x_{t_1} x_{t_2} \dots x_{t_r}$ in T where $x_{t_1} = y$, $x_{t_r} = x_k$ and $r \geq 2$ for discussion. The arguments are similar to those in Case 1 and then we can get $s_{T'}(x_k) = s_T(x_k)$.

By Case 1, 2 and 3, T' and T have the same status sequence. $\square$

Conjecture: Let T be a tree. If $s_T(x) \neq s_T(y)$ whenever $x, y \in V(T), x \neq y$ and $deg_T x, deg_T y \geq 2$, then T is uniquely determined in trees by its status sequence.

References

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